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# Abstract

In recent years, Malaysians have shown increasing interest in joining the outdoor adventure activities to relieve their stress. However, most of the individuals encounter significant challenges when browsing and booking guides. For guides, they face difficulties in communicating with explorers due to the absence of a centralized platform. The equipment providers also encounter challenges in managing their services effectively.

In this document, the introduction and related work concerning the outdoor adventure system will be outlined, followed by a motivation section that discusses the problem context, study limitations, and the novelty of the proposed research ideas. Then, the key use cases for each user role are presented, followed by the design strategies and technical requirements of the proposed system, PathFindr. Finally, a Gantt chart will be shown, illustrating the timeline of the development process of PathFindr application.

# 1.0 Introduction and Related Work

## 1.1 Introduction

In today's fast-paced digital world, more Malaysians are finding ways to relieve their stress or escape their daily routines by reconnecting with nature. Thus, outdoor activities have become an excellent and popular choice as they help improve both mental and physical well-being. Besides, outdoor experiences have also become a form of self-care, promoting both vitality and happiness.

According to the Malaysian Sports Culture Index 2022, there are 52% of Malaysians aged 13 and above are active in sports, exercise or recreational activities at least once a month (Mohd Salleh Aman et al., 2024). Besides, a survey conducted by the Department of Statistics Malaysia in 2020 also found 53.9% of population engaged in at least 1 physical activity (Yuen Mei Keng, 2023). These data reveal a growing national trend among young people toward outdoor experiences that boost physical fitness, mental health as well as personal fulfilment. In addition, the findings also highlight a cultural shift among young generation towards a more active lifestyle, where fitness and meaningful experiences are increasingly prioritized.

In order to meet this rising demand and make outdoor activity planner more accessible, PathFindr, user-friendly platform for adventure seekers. One of the key features of the software is its wide range of filters, allowing users to discover and plan activities based on their preference. As a result, users are able to enjoy a seamless and personalized planning experience that save time, as well as support informed decision making. In the meantime, once the users are ready to embark on an adventure, PathFindr provides end-to-end trip management, from booking guided tours with experienced guides and view itinerary details, to accessing equipment rental or purchase, ensuring a smooth and seamless journey.

All in all, whether you are a beginner or a seasoned explorer, PathFindr makes the experience of planning outdoor activities effortless through a seamless all in 1 platform, making adventure more accessible, organized and enjoyable.

## 1.2 Related Work

### *1.2.1 AllTrails*

AllTrails is a mobile application that designed to serve outdoor enthusiasts including hikers, trail runners and mountain bikers, by providing a comprehensive tool for exploration and navigation. This app is popular among the adventure seekers as it does provide over 450,000 trails worldwide, giving users a wide variety of options (AllTrails, n.d.). Besides, AllTrails appeals to both casual explorers and experience professionals, as it offers trails that cater to different skills levels, from beginners to advanced users. Moreover, the platform integrates advanced technology to offer a seamless experience for users to explore and enjoy their adventures.

Coming to the main features of the platform, AllTrails primarily focuses on trail finder, which enables users to search for trails based on their preferences like activity type, distance and difficulty level. In order to make the exploration more engaging, users can filter trails based on specific features and attractions such as lakes or waterfalls. Then, each of the trail is provided with details description and photos, as well as user reviews to help users make decision before embarking on their journey (Sendy.io, n.d.).

In terms of navigation, a real-time GPS tracking function is provided, allowing users to monitor and view their location on the trail. This feature helps ensure user and prevent users from getting lost or straying off course. Similar with other navigation apps such as Google Maps or Waze, AllTrails also implement turn-by-turn directions to allows users to receive visual guidance, making the navigation process more convenient and smoother (Caminiti, 2025). Meanwhile, for enhanced security, users can pin their locations and retrace back their step if necessary.

Alternatively, AllTrails uses React Native is the JavaScript framework to allows the software could operate efficiency on both iOS and Android devices. Moreover, the app also relies on AWS for cloud storage and server management, ensuring reliable data access and management. On the other hand, Google Maps API is implemented to serve real-time GPS functions, providing

precise location tracking and deliver up-to-date, accurate and real-time navigation information. Additionally, open-source nature such as OpenStreetMap and Leaflet are uses for the map visualization, ensuring a details and accurate route mapping (Nagar, 2025).

In short, AllTrails encourages outdoor exploration by promoting eco-friendly practices and supports trail conversation. In order to reduce the carbon footprint of outdoor activities, the software also encourage and helping users to discover local trails, minimizing the need for long-distance travel.

### *1.2.2 Komoot*

Komoot is another mobile application that serves as a route planner and navigation tool for outdoor activities. Similar with other outdoor activities planner, this app is also mainly designed for travellers, fitness enthusiasts or adventure seekers such as hikers, cyclists or runners, who is eager to explore new trails and looking for a user-friendly tool to plan their outdoor adventure (Komoot, n.d.-b). Additionally, Komoot do also provides fitness tracking capabilities, attracting users who prioritize convenience and efficiency (von Aulock, 2024).

The main function of the app lies on its route planning tools, which allows users to create personalized route for different activities such as hiking, cycling or jogging. Then, users can specify information such as starting point, destination and preferred route based on terrain (Searle, 2024). A voice guided turn-by-turn navigation features is also available, providing real-time and hands-free guidance during outdoor activities, ensuring users stay on the track without constantly checking their phones. Moreover, Komoot also enable users to download maps for offline usage in case they are in areas with unstable network connectivity (Komoot, 2023).

Another key functions of Komoot is its community aspect that encourage users to share their planned route, tours, tips and experience. This fosters a collaborative hub where users can follow other community members to explore their public routes, discover new places and exchange advice (Searle, 2024). Not to mention, fitness tracking capabilities are also integrated in Komoot, allowing users to keep track and monitor their performance, such as distance travelled, elevation gained, and calories burned (von Aulock, 2024).

Similar with AllTrains, Komoot relies on AWS and OpenStreetMap. AWS supports the backend infrastructure and syncing user data across devices, while OpenStreetMap is used for route generation purposes, leveraging its details mapping data to create an accurate route for users. Meanwhile, PostgreSQL has also been used as the database management system (Komoot, n.d.-a).

In a nutshell, Komoot encourages outdoor activities that foster a connection with nature, by highlighting eco-friendly trails and alternatives. Other than that, the platform also emphasizes on outdoor activities like hiking and cycling which helps in reducing the dependence on motor vehicles, supporting healthier, more sustainable transportation options.

# 2.0 Motivation

The development of PathFindr is driven by the increasing demand for structured and safe outdoor experiences in Malaysia. Outdoor adventure tourism is gaining traction, reflecting a broader global trend toward nature-based and experiential travel. With growing urbanization and a renewed interest in fitness, mental wellness, and eco-tourism, more Malaysians are engaging in hiking, jungle trekking, and other outdoor activities.

PathFindr emerges from this context, aiming to reimagine how adventurers connect with services, information, and one another in a seamless, role-specific ecosystem. By supporting digital transformation in the adventure tourism space, PathFindr not only offers convenience but also promotes safer, more efficient, and more trustworthy exploration experiences across Malaysia.

## 2.1 Problem Context

Despite the rising popularity of outdoor adventure activities, the current landscape in Malaysia remains fragmented. Users often rely on multiple third-party apps and informal social media groups such as social media groups and third-party blogs to discover trails, find guides, or rent necessary equipment.

This fragmentation not only introduces inconvenience but also increases the risk of fraud, miscommunication, and safety lapses. For instance, research shows that nearly half of Malaysian hikers lack reliable or precise trail information—highlighting a broader challenge in locating qualified guides on informal platforms (Mohd Sidi et al., 2020).

Studies have highlighted the frequency of medical incidents, such as dehydration, cold exposure, sprains, and lacerations, which often result from explorers not carrying adequate essentials—illuminating how fragmented planning leads to real-world harm (Daniel et al., 2021).

In Malaysia and similar contexts, hikers rely heavily on unverified local sources, and the absence of a centralized service makes it difficult to ensure that safety standards are uniformly met.

Furthermore, miscommunication between explorers and guides is another frequent issue. When planning relies on non-standard channels, vital details—such as emergency protocols, rendezvous points, and group coordination—are often overlooked, resulting in delays, stressful misalignment, and even abandonment of guided trips.

The absence of formal systems for tracking guide credentials, verifying equipment availability, and managing emergency responses contribute to reduced user trust and a high degree of operational inefficiency. Safety-focused studies have shown that structured systems with features like real-time group monitoring significantly reduce incident rates among outdoor excursion groups. For example, systems such as GVIDI, developed to track hikers collectively in real time, have demonstrated improved safety outcomes by alerting leaders to deviations or delays (García et al., 2023).

From the perspective of service providers, guides face inconsistent income due to lack of visibility and unpredictable booking channels, while equipment providers struggle with logistics and stock management in a decentralized ecosystem. Without integrated platforms that offer tools for inventory tracking, booking management, and customer engagement, these providers are forced to operate using inconsistent manual processes.

## 2.2 Limitations of Study

**1. Insecure Payment Gateway Integration:**

Although third-party APIs such as PayPal were considered for real world transaction integration, however, the inability to process refund functions securely poses as a critical barrier. This limitation introduces significant financial and user trust risks. Without a fully secure and certified payment gateway, the application cannot process real transactions, which restricts its usability to perform real monetary transactions reliably, hindering complete financial operations within the app.

**2. Sample Size and Testing Constraints:**

The research phase involved only nine participants across three core user roles, which limits the ability to generalize findings to a broader audience. While the qualitative insights gained were valuable, the small sample size may not reflect the diversity of behaviour and expectations across the wider Malaysian outdoor community.

**3. Device-Specific Feature Limitations**:

The app was built using Flutter for cross-platform compatibility, but features like GPS tracking, background location updates, and SOS alerts have not been fully optimized or tested across a wide range of devices. This could lead to inconsistent functionality, especially under varied environmental or hardware conditions like outdated mobile versions.

**2.3 Novelty of Research Idea**

The novelty of PathFindr lies in its ability to serve as a comprehensive, role-specific operational platform that unifies all major aspects of outdoor adventure planning, coordination, and execution into a single mobile application. While modern mobile apps often provide trail discovery, guide booking, or equipment rental features individually, PathFindr uniquely consolidates these functionalities to support the entire adventure lifecycle—specifically designed for the Malaysian context of outdoor tourism.

What distinguishes PathFindr is not just its inclusion of multiple user roles, but its delivery of dedicated functional modules for each role—explorer, guide, and equipment provider—within one integrated system. Rather than requiring users to alternate between different apps or communication platforms, PathFindr facilitates a one-stop environment where all essential planning, safety, and coordination tasks can be executed collaboratively and in real time.

For explorers, the platform offers more than route selection or booking. Its real-time trail coordination features—such as live location sharing with guides during trips, emergency SOS activation, and trip phase tracking—are tailored for high-risk outdoor activities. These features support safety and situational awareness, elements often missing in general travel or tourism apps.

For guides, PathFindr transforms the traditional role from passive listing to active trip management. It enables them to define detailed trail offerings, monitor explorer positions during excursions, issue real-time safety alerts, and manage each phase of the trip operationally within the platform. The guide interface reflects the responsibilities involved in managing live expeditions, not just client acquisition.

For equipment providers, PathFindr introduces functionality not usually present in travel applications, such as live inventory tracking, rental return status, and maintenance management.

These logistical features are built specifically for the operational realities of renting adventure gear, ensuring accountability and service quality.

Another key differentiator is PathFindr's use of Geoapify, a mapping API that focuses on nature-oriented trail recommendations. Unlike common services that return commercial or urban landmarks, Geoapify is suited for hikers and explorers seeking natural adventure locations. This context-aware mapping ensures that trail discovery remains relevant and precise.

Altogether, PathFindr's value lies in being not a marketplace, but a fully integrated digital operations platform tailored to the outdoor adventure sector. It centralizes functionality typically scattered across booking sites, logistics systems, safety tools, and communication channels. This level of specialization, modularity, and end-to-end service coordination is rarely found in mainstream applications, establishing PathFindr as a novel and practical solution to the fragmented reality of adventure planning in Malaysia.

# 3.0 Description of Process

## 3.1 Key Use Cases

### *3.1.1 Explorer*

**1. Discover and Explore Trails & Guides**

Explore has the capability to discover adventures, as well as receive recommendations for nearby adventure spots. Explorer is allowed to apply filters to the trails and view the details of the trail. After selecting a place, they can choose a guide who will lead them on the adventure. In the selection process, they are allowed to compare the guides and can view the profile and review of the guide.

**2. Save Trails to Wishlist**

After discovering a suitable adventure, explorers have the option to save the trail to their wishlist for future viewing or booking. This feature enables the users to revisit the trails they are interested in or passionate about efficiently, without the need to scroll through or search from the full list again.

**3. Book Guided Trips**

After selecting suitable trails and guides, the explorer can book the guided trip by specifying details such as group size, preferred date, and time. This approach helps to streamline the booking processes, ensuring a smooth transition from exploration to confirmation.

**4. Browse and Purchase Equipment**

For each booked trail, the system will provide explorers with recommended equipment based on the nature of the adventure. When browsing through the equipment marketplaces, which consist of various types of gear offered by equipment providers, the explorer can view the detailed product information and user reviews before deciding to purchase or rent the equipment.

**5. Participate in Trips in Real Time**

Upon booking a trail, the explorers are granted access to the current status of the trip, such as preparation, ongoing or completed phases. They can also access and review the safety precautions and specific instructions provided by the guide. For any inquiries or clarifications, they can communicate with the guide or other participants through the group chat feature.

During the trip, the explorers are able to track the real-time location data of themselves and the guide. If they encounter any issues or emergency situation, they are also allowed to send SOS alerts to the guide for immediate assistance.

**6. Provide Post-Trip Feedback**

After each trip, the explorers are allowed to provide feedback that serves as a reference for other users. They can rate the guide and the equipment purchased or rented from equipment providers using both star and text formats. After submitting their ratings, they are able to view the reviews they have provided for future reference.

**7. Top Up Wallet and Make Payments**

To facilitate the guide booking and equipment purchasing processes, the explorers are allowed to top-up their in-app wallet. They are also allowed to view their payment history, either in graph or text form.

**8. Manage Profile, Account and Settings**

The explorers are allowed to manage their profile, including updating their profile picture, account password, and other personal information. This ensures that their account remains up to date. Besides that, explorers are also allowed to adjust localization settings and application brightness, ensuring a personalized and comfortable user experience.

### *3.1.2 Guide*

**1. Manage Offerings and Availability**

Guides can configure their availability using the calendar interface and input fields provided within the system. They are also able to view their scheduled guidance activities through an integrated calendar view. When creating a trip offering, guides can define key trip details such as difficulty level and estimated duration. Additionally, they can set the pricing and participant capacity for each trip, which facilitates better planning and management of their guiding services.

**2. Handle Bookings and Communication**

After an explorer makes a booking, the system will notify the respective guide and grants them access to manage the booking. Guides have the option to accept or reject the trip based on their availability and preferences. During the trip preparation phase, they are allowed to provide safety guidelines and manage the list of participants. Additionally, guides can access emergency contact information of the explorers to ensure appropriate response measures are in place if needed. To facilitate pre-trip coordination, guides can also communicate with explorers through the integrated group chat feature.

**3. Lead and Monitor Trips**

For each accepted trip, the guides will be granted full access to monitor and manage the progress of the trip. They can perform various operations, including updating the trip phase and issuing safety alerts when necessary. During the trip, the guides can track the real-time location of all participants via an integrated map interface, which helps ensure the safety of each member and prevents potential accidents. If a participant sends an SOS alert, the system will notify the guide immediately with an emergency sound prompt, allowing for a prompt and effective response.

**4. Maintain Reputation and Earnings**

After completing a trail and receiving feedback from explorers, the guide will be notified by the system and can access the ratings and comments provided. By reviewing the summarized

information such as average ratings and feedback history, the guides can identify the areas for improvement and enhance the quality of their services. In addition, they also allowed to update their profile, including their photo and personal description, thereby presenting a more professional image an effectively promote their offerings.

**5. Withdraw Received Payments**

Earnings from each guided tour are credited directly to the in-app wallet of the guide. For guides, they are allowed to request withdrawals of their income to their bank account. In addition, they also can view the detailed transaction history, which includes records of completed tours, withdrawal activities, and overall earnings. The system will also analyze their earnings and provide them with valuable insights.

### *3.1.3 Equipment Provider*

**1. Manage Equipment and Inventory**

The equipment providers have the capability to manage the items offered through their store. For instance, they are allowed to add new equipment for both rental and purchase as new stock becomes available. For each item, providers are granted access to monitor stock levels and track the status of the item, such as available, rented or under maintenance. Additionally, they are also allowed to organize the equipment into relevant categories, including tents, hiking poles and other gear types.

**2. Handle Orders and Logistics**

Once an explorer places an order, whether for rental or purchase, the equipment provider will be notified by the system, ensuring timely processing of the request. Since the explorers may choose either pickup or delivery options, equipment providers are required to manage the chosen logistics accordingly. If the explorer selects the delivery option, the equipment provider will need to update the logistics status to reflect stages such as “Preparing” and “Shipped Out”. For the rental orders, the equipment providers are required to handle the return process, including verifying the condition of the returned items and updating the inventory status.

**3. Promotions and Customer Engagement**

The equipment providers are allowed to bundle multiple items into promotional packages and offer discounted prices as a strategy to increase sales and attract customers. Additionally, the system also allows the explorers to leave reviews after purchasing or renting the equipment. As a result, the equipment providers can improve their products and enhance overall service quality by reviewing this feedback.

**4. Communicate with Explorer**

If explorers encounter any issues related to the equipment or services provide by equipment providers, they can use the integrated chat feature to initiate direct communication. Through this

channel, the equipment providers can clarify and provide the information about the equipment, as well as share detailed insights, thereby fostering better understanding and improving customer satisfaction.

**5. Withdraw Received Payments**

For each completed rental or purchase, the corresponding earnings will be credited to the in-app wallet of the equipment provider. The equipment providers can access a summarized breakdown of their income, with filtering options by month and year. These data will be analyzed by the system automatically to offer actionable financial insights. They also can request a withdrawal of their earnings, which will be transferred to their bank account.

## 3.2 Design Strategies

### *3.2.1 UI and Visual Design Strategy*

1. **Logo**

   **Figure 1**

   *PathFindr System Logo*

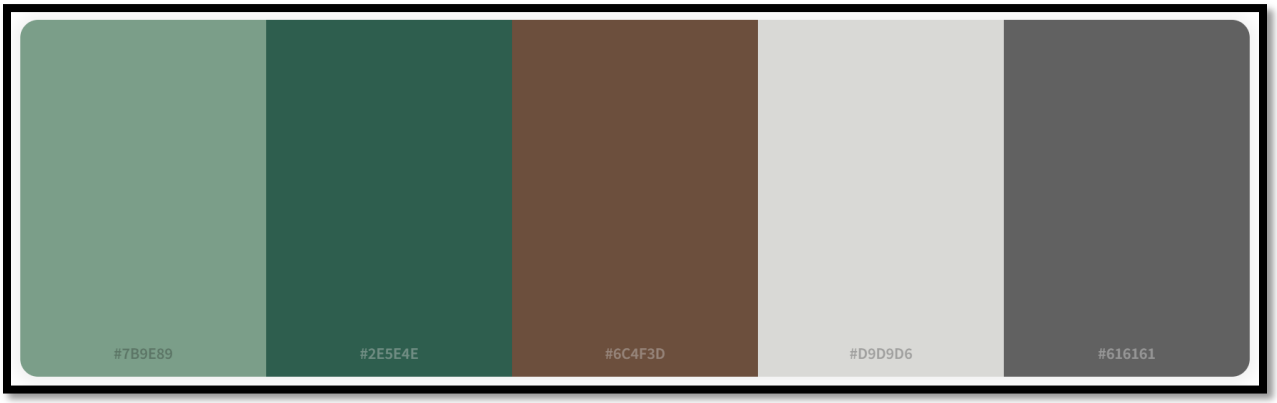



The diagram shows the logo of the PathFindr system which a combination of the system name, PathFindr, a mountain that represents the main domain of the system, outdoor activities and a slogan: Gear Up. Head Out. Explore More.

2. **Colour Palette**

   **Figure 2**

   *PathFindr System Color Palette*



The diagram above shows the color palette used in our system, which consists of 5 colors. A green color theme is used for PathFindr as it represents nature, growth, and outdoor exploration, which are the key elements of the system.

## 3. Wireframe

### a. Main Page

**Figure 3 & 4**

*Main Dashboard*




The wireframe above shows the main page of PathFindr, where users can click a button to access the login page. Then, when the user scrolls down, they can view general information about the PathFindr system.

**b. Login Page**

**Figure 5**

*Login Page*

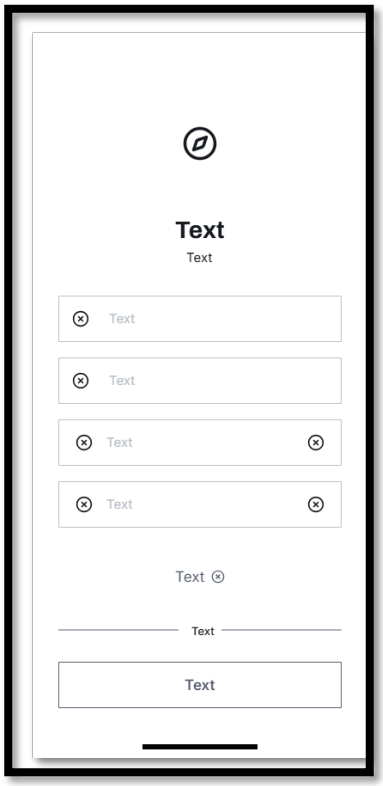



The wireframe above shows the login page of the system, where users are given 2 options to login to. The first option is using a registered email and password. On the other hand, users can choose to sign in with their Google account.

**c. User Registration Page**

**Figure 6**

*Registration Page*

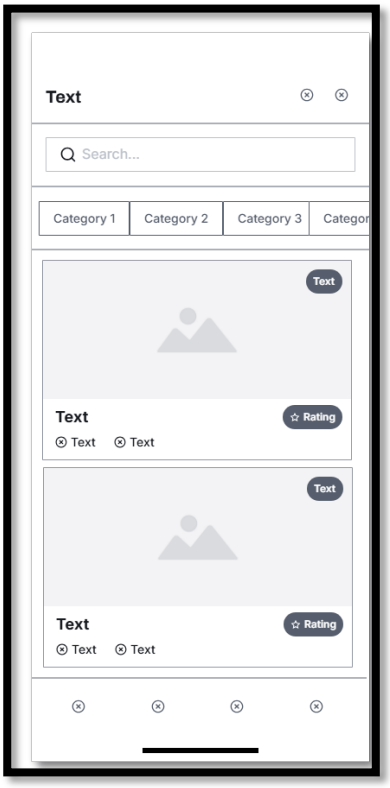



The wireframe above shows the registration page, where users can create a new account in order to access to the system. Basic information is needed, after which users can click a button to proceed to the next steps. An alternative method is also provided, allowing users to create a PathFindr account by signing in with their Google account.

**d. Explorer Main Page**

**Figure 7**

*Explorer Role - Dashboard Page*

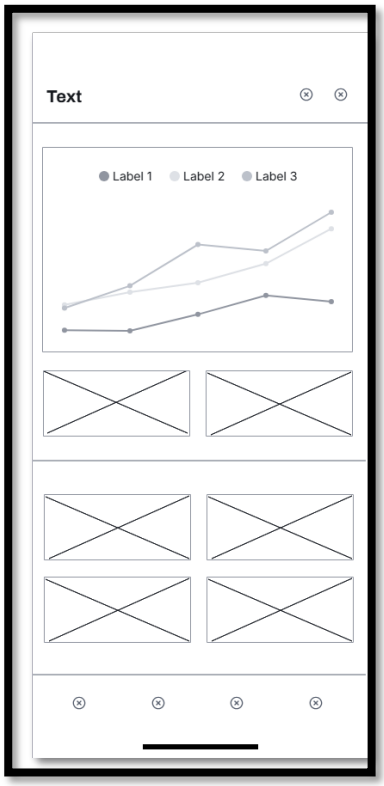



The wireframe above shows the main dashboard of the Explorer, which is divided into multiple sections. Starting off with the top right corner, users can view notifications and access their profile. Then, a search bar and filter function are provided for user convenience. A list of outdoor activity is then displayed based on the applied filtration. Lastly, a navigation bar is also available to allow users to access to other functionalities.

**e. Guide Main Page**

**Figure 8**

*Guide Role - Dashboard Page*

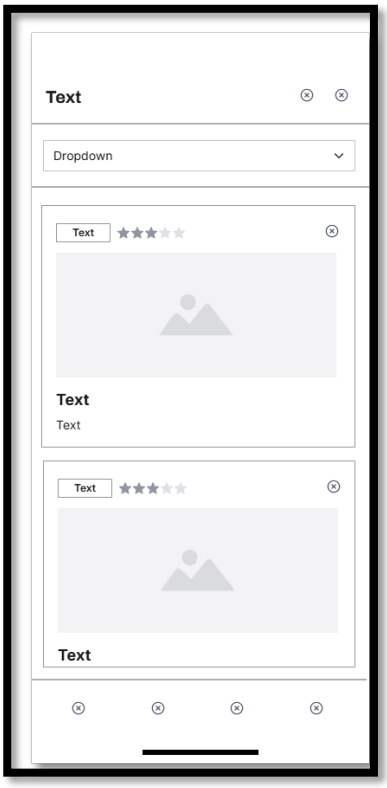



The wireframe above shows the dashboard for the Guide. Similar to the Explorer's, users can find the notification and profile button at the top right of the page. Then, a graph is displayed to show their revenue. Multiple buttons are also provided as shortcuts for users to access other functions. Not to mention, a bottom navigation bar is also provided for user convenience.

**f. Equipment Provider Main Page**

**Figure 9**

*Equipment Provider Role - Dashboard Page*




The wireframe above shows the main dashboard of Equipment Provider. It shares a similar design with Explorer and Guide dashboard, including the notification and profile button at the top right. Then, users are able to filter equipment based on the status, and the results are displayed accordingly. To enhance the user experience, a navigation bar is also provided for access to other functionalities.

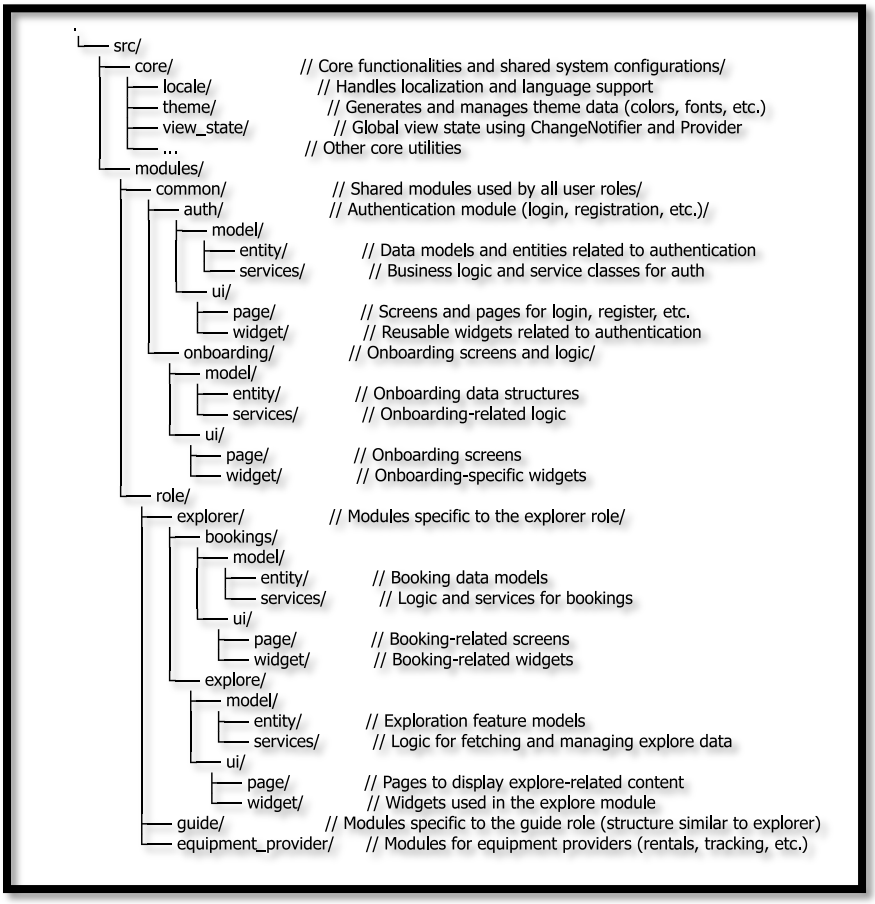
### *3.2.2 System and Code Architecture Design*

#### 1. MVVM Architecture

The Model-View-ViewModel (MVVM) architecture is adopted in the development of the PathFindr application to ensure a clear structural separation between the user interface, business logic, and data management. This architectural pattern promotes modularity, allowing individual components or modules to operate independently without affecting others. As a result, issues within one module can be addressed or debugged without disrupting the overall system, thereby enhancing maintainability, scalability, and development efficiency. The file structure of the Flutter source code is presented below:

**Figure 10**

*File Structure of Flutter Source Code*



## 3.3 Technical Requirements

### *3.3.1 Tools and Simulation Platforms Used*

1. **Visual Studio Code**

**Figure 11**

*Visual Studio Code Logo*




Visual Studio Code, a lightweight and open-source IDE, is used as the main development tool for the PathFindr system. This tool is chosen as it consists of various useful extensions, including Flutter and Dart support, Git integration and intelligent code completion, as well as an error checker. These features allow the development team to develop and maintain the system in an efficient and effective manner.

2. **Android Studio**

**Figure 12**

*Android Studio Logo*



Android Studio is capable of simulating Android devices within a desktop environment. It includes access to the Android Software Development Kit (SDK) and related tools, which enable the development team to connect with virtual and physical Android devices, as well as to test and debug applications on these devices.

### 3. Xcode

**Figure 13**

*Xcode Logo*




Since the PathFindr application aims to deploy on the iOS platform, the use of Xcode enables the developers to develop and debug the application using both iOS simulator and physical iPhone devices. As the configuration and development environment for iOS differ significantly from Android, it is crucial and essential to include iOS-specific development to ensure compatibility and consistent performance across both platforms.

### *3.3.2 Backend and Storage*

1. **Firebase**

   **Figure 14**

   *Firebase Logo*




Firebase is a cloud-based platform provided by Google that offers a suite of backend services for application development. The primary reason of integrating Firebase as the backend and storage for the PathFindr application is its strong compatibility with Flutter, which simplifies development and accelerates deployment. Moreover, Firebase provides advanced features such as real-time database, user authentication, and cloud storage at a low cost. This approach is essential as it enables the development team to manage the user data, as well as maintain a scalable backend infrastructure efficiently with minimal setup and maintenance.

### *3.3.3 Third-Party API Integration*

1. **Google Maps API**

   **Figure 15**

   *Google Maps API Logo*



The Google Maps API is used to provide map visualization of specific locations and routing features within the PathFindr application. This integration is essential as it allows users to accurately identify the location of adventure sites and understand how to reach them. The use of the Google Maps API is primarily due to its regularly updated data and high reliability, ensuring that users receive accurate and current geographical information during their trip planning process.

**2. Geoapify**

**Figure 16**

*Geoapify Logo*




Compared to the Google Maps API, the Geoapify API is more capable of providing nature-oriented locations for exploration, such as hiking trails, caves, and outdoor landmarks, rather than general tourist attractions like museums, which are commonly returned by Google Maps API. This approach is essential as it helps prevent the users from being confused by irrelevant information within the application, ensuring a more focused and context-aware experience tailored to outdoor adventure planning.

**3. Open-Metro API**

**Figure 17**

*Open-Metro API Logo*

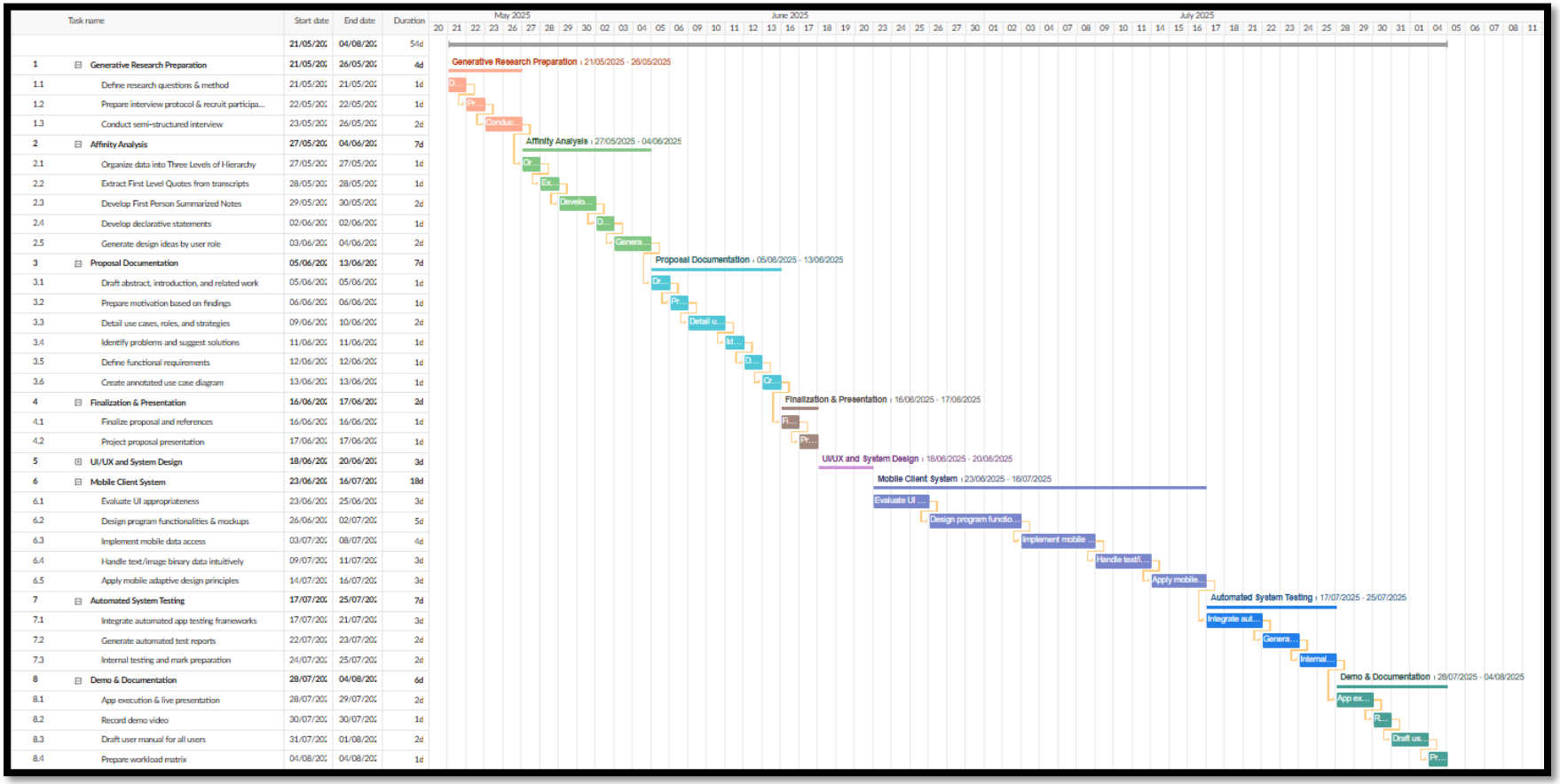



Since the PathFindr applications includes a feature that allows the explorers to check weather conditions before booking for a tour, the Open-Metro API contributes significantly to supporting this functionality. Its fast and accurate weather forecast data enables the users to access reliable environmental information, which helps them to make safer and more informed decisions when planning their outdoor adventures.

# 4.0 Tasks and Milestone (Gantt Chart)

**Figure 18**

*Gantt Chart for the PathFindr System Development Process*



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